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SLOT-C

1. Intelligent Agents in smart Home Automation

Introduction

An intelligent agent observes its environment using sensors and takes actions to achieve a goal. In a smart home, it control lights, fans air conditioners, and security systems.

1. Simple Reflex Agent

→ works based on current input

→ follows simple condition-action rules

Example: If motion is detected, the light turns on

Advantage: Fast and simple

Disadvantage: cannot rember past events.

2. model-Based agent

uses current input and past information

maintains an internal state of the environment

example: Remembers wheather a person is inside a room.

Advantage: makes better decisions using stored information

Disadvantage: Requires memory

3. Goal Based agent

Takes actions to achieve a specific goal

plans actions before making decisions.

example: switches on the AC before the user

arrives home

→ Advantage : helps achieve desired results.

→ Disadvantage : planning takes more time.

4. Utility - Based agent.

⇒ Selects the best action by comparing different choices

→ considers comfort, energy usage, and security

→ Example : adjusts the AC to save electricity while maintaining comfort

→ advantage : Balances multiple needs

→ disadvantage : more complex to design.

Best approach :

A hybrid intelligent agent combining model-based, goal-based, and utility-based agents is best. A learning component can understant user preferences over time, while simple reflex rules handle urgent events such as security alarms.

Conclusion : A hybrid agent provides comfort, saves energy, improves security, and learn user preferences.

2. Solving an unknown maze using ucs and IDS

Introduction

A robot in an unknown maze must explore different paths to find the exit. uniform cost search (ucs) and Iterative Deepening search (IDS) are uninformed search algorithms.

1. uniform cost search (ucs)

Explores the path with the lowest total cost first

uses a priority queue

Finds the least cost path when movement costs are negatives.

advantage: Finds the cheapest path.

Disadvantage: uses more memory and may take more time.

2. Iterative Deepening search (IDS):

Search with a depth limit

Increases the limit step by step until it find the exit

finds the shortest path in number of steps when all moves have equal cost.

Advantage: uses less memory

Disadvantage: Repeats some searches

Feature	UCS	IDS
Time	can be high	$O(b^d)$
Space	High	low, $O(bd)$
Complete	Yes, under standard cost conditions	Yes, for finite branching
Best for	Different movements	Equal-cost

Here b is the number of possible next moves and d is the depth of the exit.

Types of Intelligent agents in the maze

1. Simple reflex agent : moves according to current sensor inputs

2. model Based agent : Remembers visited places and obstacles.

3. goal Based agent : chooses actions to reach the exit.

4. utility Based agent : chooses path Based on cost, time, and safety.

Best approach :

A model-based and goal-based agent using ucs is best when the maze has different movement costs

→ The model based agent rembers obsteales
→ the goal based agent focuses on reaching the exit
→ ucs finds the least cost path
→ The robot updates its map as it explores the unknown maze.

If all moves have equal costs and memory is limited, IDS is a good choice.

Condusion : ucs finds the least cost path, while IDS saves memory. A model based, goal beused agent helps the robot explore the maze and reach the exit efficency.